

Engineering Reboot 2026

Week-02 Day-06 Engineering Lab Manual v4.0 - Part 2 Laboratory Exercises

Laboratory Exercise Overview

You will create five Python programs:

high_temperature_alert.py
temperature_status.py
age_classification.py
grade_evaluation.py
decision_program.py

All files reside under:

Week-02\Exercises\Day-06\

All exercises use NUMERIC INPUT via input() and int().

Exercise 01 - High Temperature Alert

Filename:

high_temperature_alert.py

Input Method:

NUMERIC INPUT

Implementation:

```
temperature = int(input("Enter temperature: "))
```

```
if temperature > 80:
```

```
    print("Air conditioning recommended")
```

Example Console Session:

Enter temperature: 85

Air conditioning recommended

Evidence:

high_temperature_alert-Code.png

high_temperature_alert-ExpectedResult.png

Exercise 02 - Temperature Status Evaluation

Filename:

temperature_status.py

Implementation:

```
temperature = int(input("Enter temperature: "))
```

```
if temperature > 80:
```

```
    print("Air conditioning recommended")
```

```
else:
```

```
    print("Temperature acceptable")
```

Validation Case A:

Enter temperature: 65
Temperature acceptable

Validation Case B:

Enter temperature: 90
Air conditioning recommended

Evidence:
temperature_status-Code.png
temperature_status-ExpectedResult.png

Exercise 03 - Age Classification

Filename:
age_classification.py

Implementation:
age = int(input("Enter age: "))

```
if age >= 18:  
    print("Adult")  
else:  
    print("Minor")
```

Validation Case A:

Enter age: 16
Minor

Validation Case B:

Enter age: 21
Adult

Evidence:
age_classification-Code.png
age_classification-Minor.png
age_classification-Adult.png

Exercise 04 - Grade Evaluation

Filename:
grade_evaluation.py

Implementation:
score = int(input("Enter score: "))

```
if score >= 90:  
    print("Grade A")  
elif score >= 80:  
    print("Grade B")  
elif score >= 70:  
    print("Grade C")
```

```
else:  
    print("Needs Improvement")
```

Validation Example:

Enter score: 83
Grade B

Evidence:
grade_evaluation-Code.png
grade_evaluation-ExpectedResult.png

Exercise 05 - Decision Program

Filename:
decision_program.py

Implementation:
voltage = int(input("Enter voltage: "))

```
if voltage < 110:  
    print("Voltage Low")  
elif voltage > 120:  
    print("Voltage High")  
else:  
    print("Voltage Normal")
```

Validation Matrix:

Input: 100
Output: Voltage Low

Input: 115
Output: Voltage Normal

Input: 125
Output: Voltage High

Evidence:
decision_program-Code.png
decision_program-Voltage-Low.png
decision_program-Voltage-Normal.png
decision_program-Voltage-High.png

Troubleshooting Guidance

Common issues:

- ValueError caused by entering non-numeric input.
- Missing indentation.
- Using '=' instead of '=='.
- Running the wrong Python file.
- Forgetting to save changes before execution.

Verify inputs carefully and re-run validation.

Completion Criteria

Part 2 is complete when:

- All five programs execute successfully.
- Validation cases pass.
- Evidence screenshots are captured.
- Files are stored in the correct repository location.